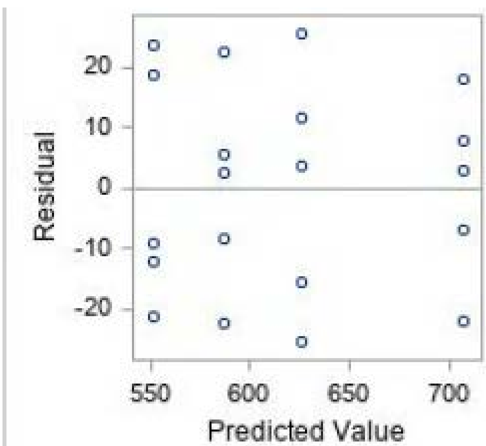
The Steps in Verifying the Validity of the Underlying Assumptions for the ANOVA Models

In Handout 3, the approach for verifying the underlying assumptions for ANOVA models, based on various types of residuals, have been discussed. We summarize this here and illustrate on the Plasma Etching Example 3.1, that has been discussed in Section 3.1 in the textbook.

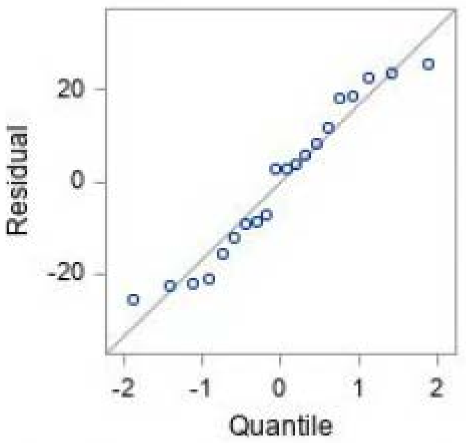
The SAS program and the SAS output for this data set is posted in Husky CT. First we examine the plot of the raw residuals vs. predicted values that is posted below:



When examining this plot, one does not want to see any visible patterns. The points should be scattered randomly around the 0 line, which is clearly the case here for our data set. This implies that the linear ANOVA model is ok and that the variances of the random errors are fixed.

If we would have seen in the plot above that there is a clear increasing (decreasing) triangular pattern then we would say that the variances of the random errors are increasing (decreasing) as the predicted value are increasing. This will imply that the assumption of fixed variances for the random errors is violated.

Next we examine the normality assumption for the random errors. First we examine the normal probability plot, the Q-Q plot for this data set given below:



When examining this plot, we would like to see a straight-line pattern, without strong curvature or long parallel line to the horizontal axis. This is the case in the plot above, which implies that most likely the normality assumption for the random errors is valid. Sometimes it will not be easy to decide based on the normal probability plot if the normality assumption is valid. Therefore, we also use the W statistic and its p-value to assess the normality assumption. From the PROC UNIVARIATE NORMAL; output we get the following:

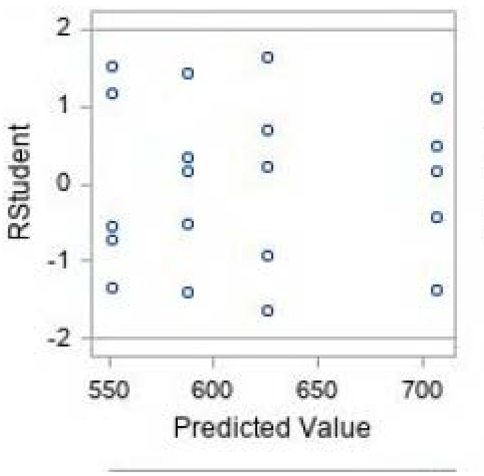
Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.93752	Pr < W	0.2152
Kolmogorov-Smirnov	D	0.111872	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.051535	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.375887	Pr > A-Sq	>0.2500

We see that the W statistic has a p-value of .2152 > .10.

Our "rule of thumb" states that as long as the p-value is greater than .10 we will confirm that the normality assumption for random errors is valid. If it is less than .01 then we will say that it is violated. If the p-value of the W statistic is greater than .01 but less than .10, than we will say that it is inconclusive. If it is close to .10 one may still accept the normality assumption.
For our data set, we can conclude that the normality assumption is valid.

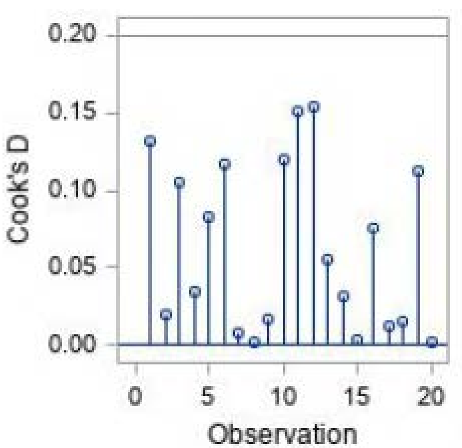
Next we examine the plots in the Fit Diagnostics panel in the SAS output for outliers and influential observations.

To detect outliers, we examine the plot of RSTUDENT (jackknifed) residuals vs. predicted values, that is presented below:



The rule we are using is: any observation with an RSTUDENT (jackknifed) residual that in absolute value is greater than 4 is declared as an outlier. Since all RSTUDENT values in the plot are greater than -2 and less than 2, there are no outliers in this data set.

For influential observations we examine the Cook's D values in the following plot:



Since all Cook's D values are less than 1, we conclude that there are no influential observations in this data set.

Remarks. 1. If the normality assumption or constant variance assumption for the random errors in the model is violated, than we will transforms the response variable. The log transformation is sometimes helpful. See Lecture Notes no 5 for more details.

2. If we detect an outlier or an influential observation, we might have to remove it from the data set if it contributes to the violation of one of the underlying assumptions.